# AWS VPC Peering & Secure Cross-VPC Architecture

*Hands-On Implementation, Routing Logic & Verification Report*


**Project:** AWS Multi-VPC Architecture & Peering
**Environment:** Amazon Web Services (AWS)
**Author:** **Megha Singh (AWS Internship Batch 1)**
**Core Protocols:** IPv4, CIDR, SSH (ProxyJump), ICMP


## 1. Theoretical Overview & Cloud Architecture

Amazon Virtual Private Cloud (Amazon VPC) provides isolated virtual networks within the AWS cloud where compute resources, databases, and network gateways operate under custom-defined IP address spaces. In enterprise environments, multi-VPC architecture is deployed to separate distinct operational environments (such as Development, Staging, and Production) or different organizational units.

### Key Architectural Concepts

- **VPC Peering:** A networking connection between two VPCs that enables routing of traffic using private IPv4 or IPv6 addresses. Instances in either VPC communicate as if they are within the same network without using public internet gateways, NAT devices, or VPN connections.
- **Non-Overlapping CIDR Blocks:** VPC Peering strictly requires that connected VPCs do not have overlapping Classless Inter-Domain Routing (CIDR) IP ranges. In this project, VPC A uses 10.10.0.0/16 and VPC B uses 10.20.0.0/16.
- **Bastion Host (Jump Box):** A hardened, special-purpose server located in a public subnet configured with strict firewall rules. It serves as the single access point to bridge internal administrative SSH management into private subnets.
- **Stateful Security Groups:** Virtual firewalls controlling inbound and outbound traffic at the ENI (Elastic Network Interface) level. Inbound traffic allowed automatically permits outbound responses regardless of outbound rules.
- **Custom Route Tables:** Tables containing rules (routes) that determine where network traffic from subnets or gateways is directed. Active peering requires explicit routes pointing destination CIDRs to the peering connection ID (pcx-...).

## 2. Architecture Topology Summary

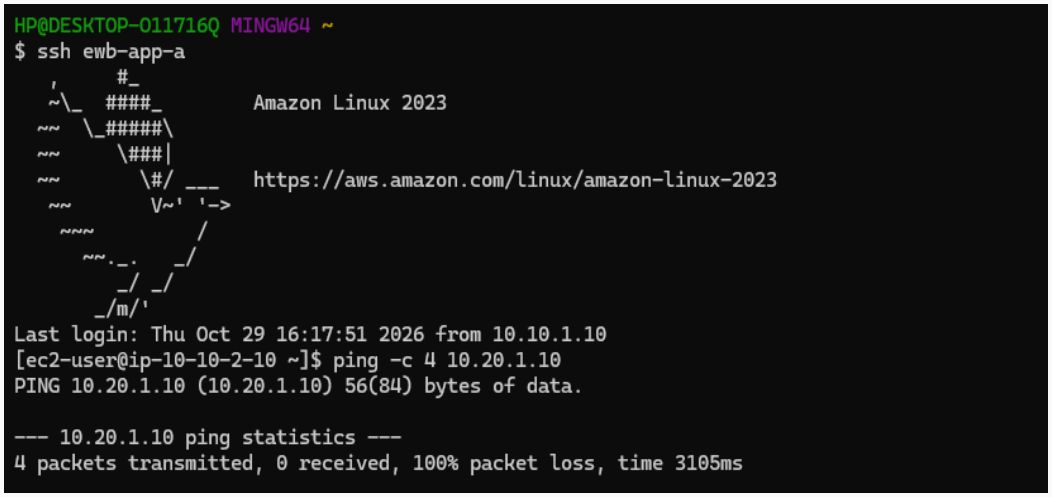
| Resource Name | VPC / Network | Subnet CIDR | IP Assignment & Type |
|---|---|---|---|
| ewb-bastion | VPC A (10.10.0.0/16) | 10.10.1.0/24 (Public) | Public IP & Private 10.10.1.X |

| | | | |
|---|---|---|---|
| ewb-app-a | VPC A (10.10.0.0/16) | 10.10.2.0/24 (Private) | Private IP 10.10.2.10 |
| ewb-app-b | VPC B (10.20.0.0/16) | 10.20.1.0/24 (Private) | Private IP 10.20.1.10 |

## 3. Hands-On Lab Tasks & Screenshot Evidence

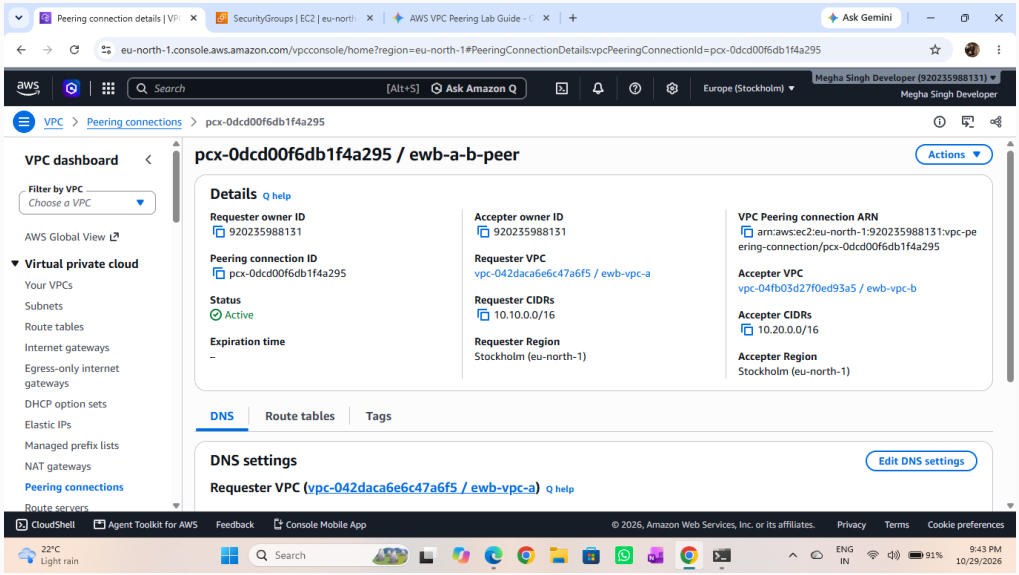
### Task 1: Baseline Connectivity Failure (Before Peering)

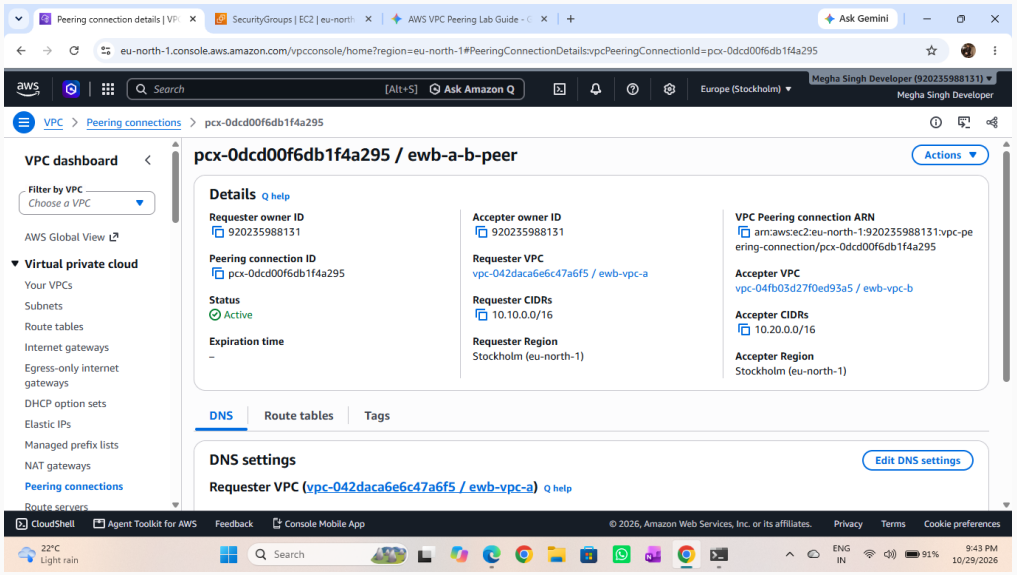
Before configuring the VPC Peering connection and custom route table entries, an initial baseline test is conducted from ewb-app-a (VPC A) targeting ewb-app-b (10.20.1.10 in VPC B). Because no route exists between the isolated VPCs, the network traffic has no gateway path, resulting in 100% packet loss.




### Task 2: VPC Peering Connection Creation & Active Status

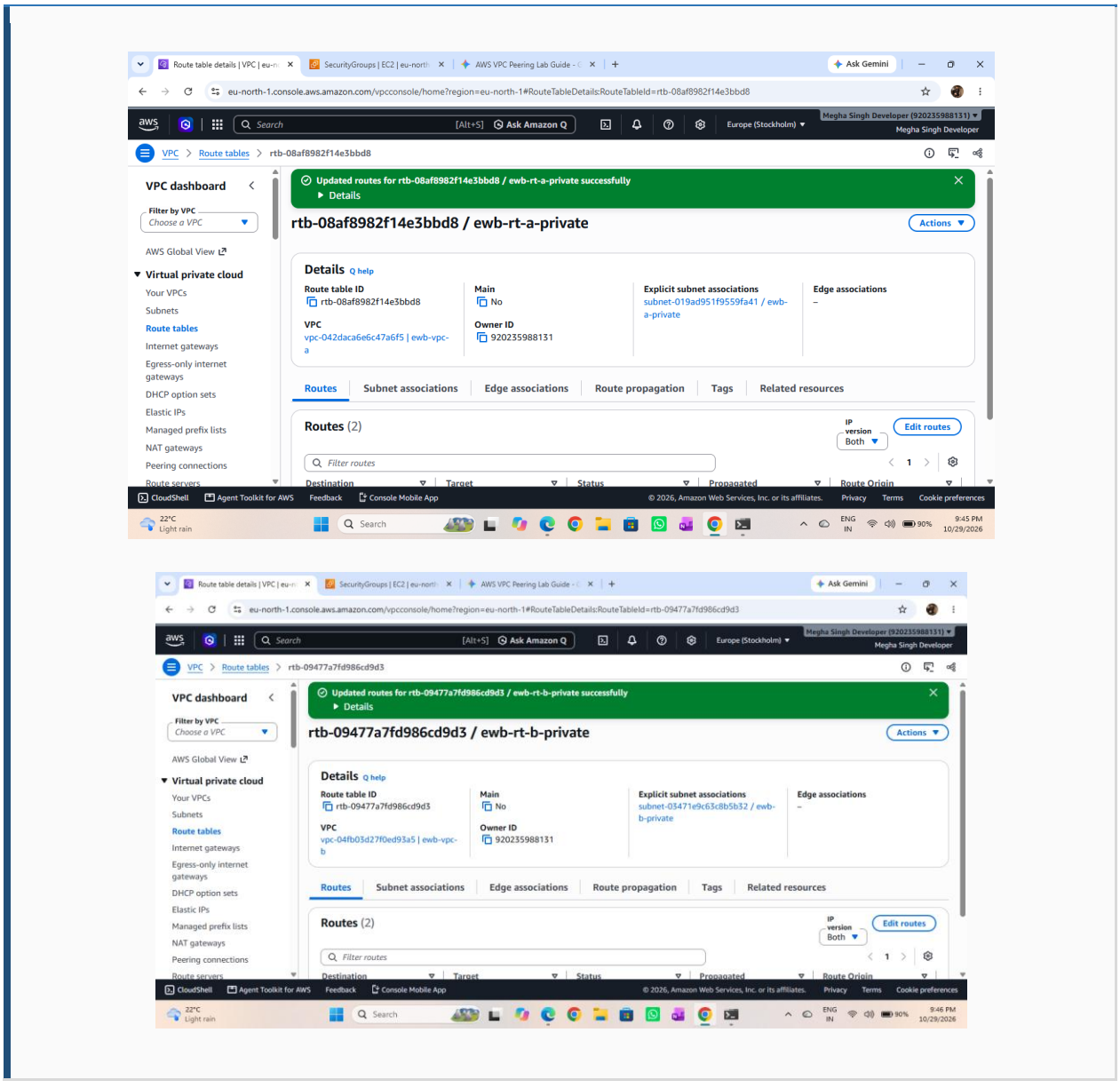
A VPC Peering Connection (ewb-a-b-peer) is requested from requester VPC A (10.10.0.0/16) to accepter VPC B (10.20.0.0/16). Once accepted in the AWS VPC Management Console, the peering status transitions to Active.

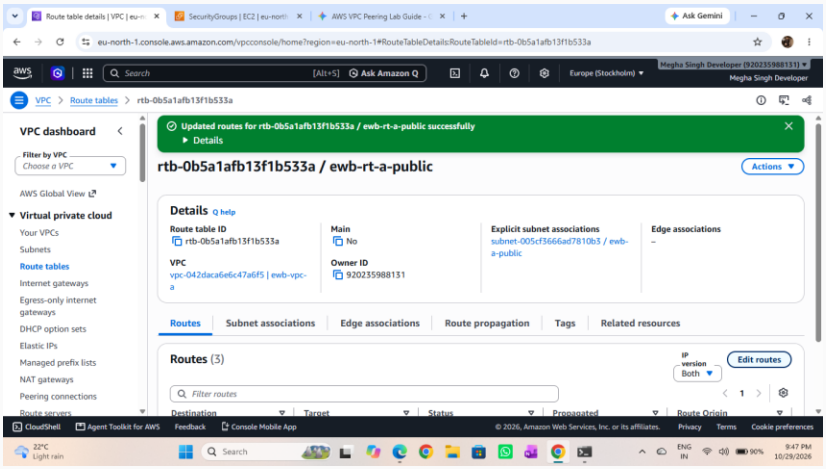

Peering connection details | VPC
SecurityGroups | EC2 | eu-north-1
AWS VPC Peering Lab Guide - C
Ask Gemini
eu-north-1.console.aws.amazon.com/vpcconsole/home?region=eu-north-1#PeeringConnectionDetails:vpcPeeringConnectionId=pcx-0dcd00f6db1f4a295
Search
[Alt+S]
Ask Amazon Q
Europe (Stockholm)
Megha Singh Developer (920235988131)
Megha Singh Developer
VPC > Peering connections > pcx-0dcd00f6db1f4a295
VPC dashboard
Filter by VPC
Choose a VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
pcx-0dcd00f6db1f4a295 / ewb-a-b-peer
Actions
Details
Requester owner ID
920235988131
Peering connection ID
pcx-0dcd00f6db1f4a295
Status
Active
Expiration time
–
Accepter owner ID
920235988131
Requester VPC
vpc-042daca6e6c47a6f5 / ewb-vpc-a
Requester CIDRs
10.10.0.0/16
Requester Region
Stockholm (eu-north-1)
VPC Peering connection ARN
arn:aws:ec2:eu-north-1:920235988131:vpc-peering-connection/pcx-0dcd00f6db1f4a295
Accepter VPC
vpc-04fb03d27f0ed93a5 / ewb-vpc-b
Accepter CIDRs
10.20.0.0/16
Accepter Region
Stockholm (eu-north-1)
DNS
Route tables
Tags
DNS settings
Edit DNS settings
Requester VPC (vpc-042daca6e6c47a6f5 / ewb-vpc-a)
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
22°C
Light rain
Search
ENG
IN
91%
9:43 PM
10/29/2026


Peering connection details | VPC
SecurityGroups | EC2 | eu-north-1
AWS VPC Peering Lab Guide - C
Ask Gemini
eu-north-1.console.aws.amazon.com/vpcconsole/home?region=eu-north-1#PeeringConnectionDetails:vpcPeeringConnectionId=pcx-0dcd00f6db1f4a295
Search
[Alt+S]
Ask Amazon Q
Europe (Stockholm)
Megha Singh Developer (920235988131)
Megha Singh Developer
VPC > Peering connections > pcx-0dcd00f6db1f4a295
VPC dashboard
Filter by VPC
Choose a VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
pcx-0dcd00f6db1f4a295 / ewb-a-b-peer
Actions
Details
Requester owner ID
920235988131
Peering connection ID
pcx-0dcd00f6db1f4a295
Status
Active
Expiration time
–
Accepter owner ID
920235988131
Requester VPC
vpc-042daca6e6c47a6f5 / ewb-vpc-a
Requester CIDRs
10.10.0.0/16
Requester Region
Stockholm (eu-north-1)
VPC Peering connection ARN
arn:aws:ec2:eu-north-1:920235988131:vpc-peering-connection/pcx-0dcd00f6db1f4a295
Accepter VPC
vpc-04fb03d27f0ed93a5 / ewb-vpc-b
Accepter CIDRs
10.20.0.0/16
Accepter Region
Stockholm (eu-north-1)
DNS
Route tables
Tags
DNS settings
Edit DNS settings
Requester VPC (vpc-042daca6e6c47a6f5 / ewb-vpc-a)
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
22°C
Light rain
Search
ENG
IN
91%
9:43 PM
10/29/2026

## Task 3: Custom Route Table Updates

To establish bi-directional routing, explicit route rules are added to the route tables. VPC A's route table is updated with destination 10.20.0.0/16 pointing to pcx-..., and VPC B's route table is updated with destination 10.10.0.0/16 pointing to the same pcx-... ID.






## Task 4: Successful Cross-VPC ICMP Ping Verification

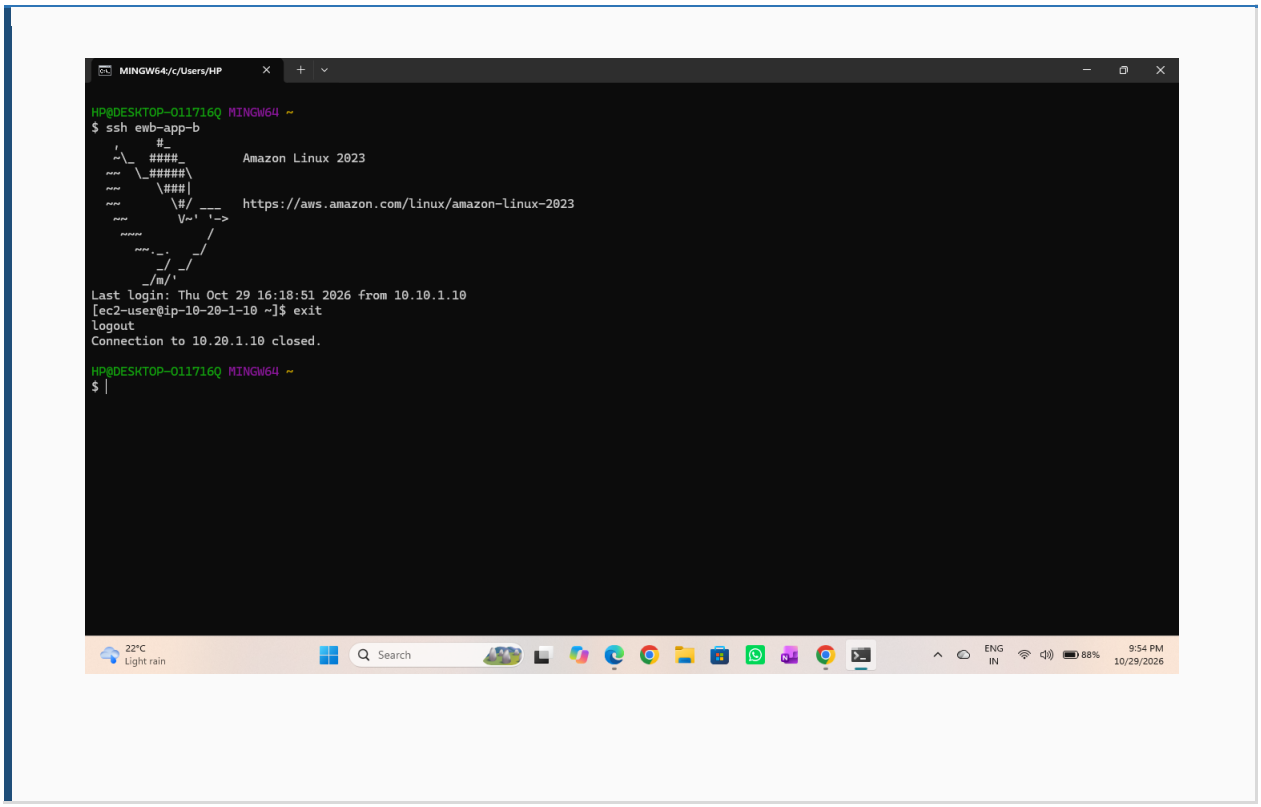
With active peering and routing entries established, an ICMP echo request is executed from ewb-app-a to ewb-app-b (10.20.1.10). The ping test verifies 0% packet loss, confirming active low-latency private connectivity.

```
HP@DESKTOP-O11716Q MINGW64 ~
$ ssh ewb-app-a
        #_
    ~\_  ####_        Amazon Linux 2023
   ~~  \_#####\
   ~~     \###|
   ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
    ~~       V~' '->
     ~~~         /
       ~~._.   _/
          _/ _/
        _/m/'
Last login: Thu Oct 29 16:10:02 2026 from 10.10.1.10
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.550 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.297 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.154 ms
64 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.169 ms

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3120ms
rtt min/avg/max/mdev = 0.154/0.292/0.550/0.158 ms
[ec2-user@ip-10-10-2-10 ~]$
```

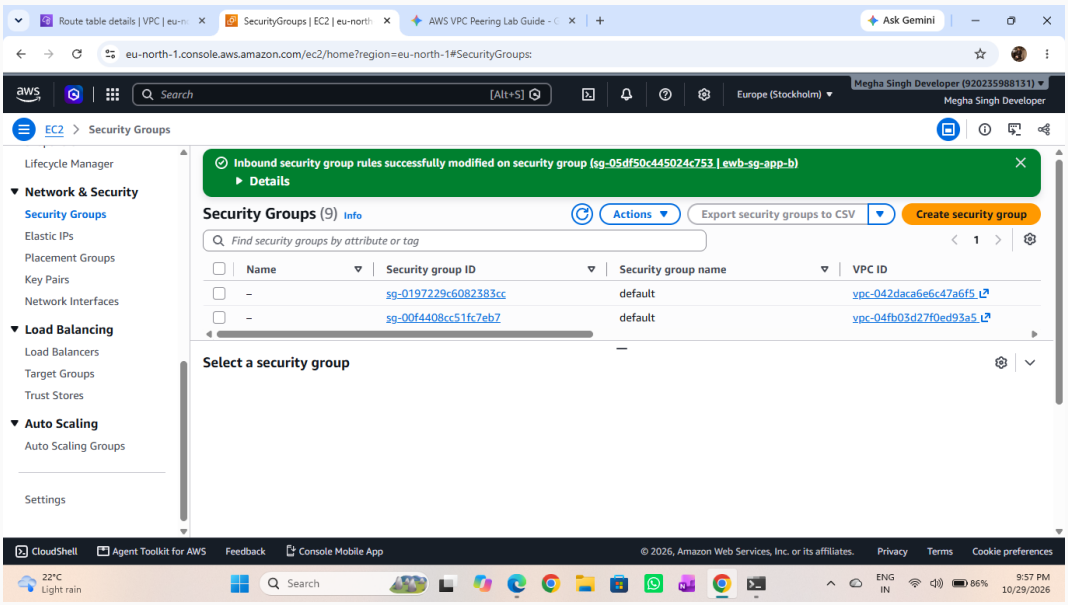
## Task 5: Private SSH Access via Bastion (ProxyJump)

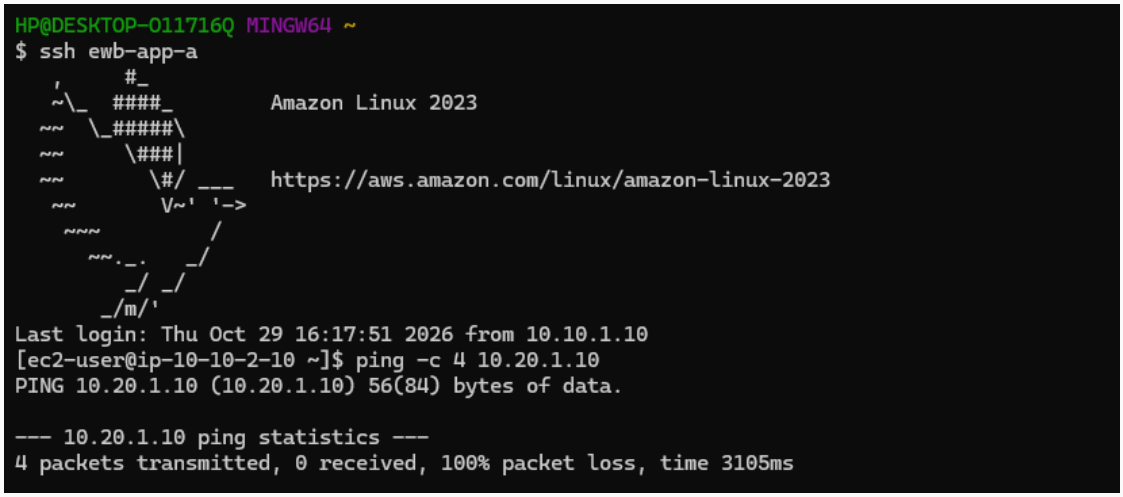
Using local SSH client configuration (~/.ssh/config) with ProxyJump enabled, SSH access is established directly from the administrator's local machine to private instance ewb-app-b through ewb-bastion without storing private keys on intermediate servers.




## Task 6: Controlled Failure Test (Security Group Rule Revocation)

To prove that Security Groups actively enforce stateful packet filtering, the ICMP Echo Request inbound rule is temporarily removed from ewb-sg-app-b. Re-running the ping test from ewb-app-a confirms immediate traffic block.




## Task 7: Resource Cleanup & Termination

To adhere to cloud cost management standards and avoid unexpected AWS billing, all EC2 instances (bastion, app-a, app-b) were terminated, peering connections deleted, and custom VPC infrastructure completely removed.

## 4. Conclusion & Learning Outcomes

This lab successfully demonstrated the end-to-end design, implementation, verification, and teardown of an AWS multi-VPC architecture. Key competencies gained include configuring non-overlapping IP spaces, managing peering routing, securing private subnets via Bastion ProxyJump tunneling, and validating stateful firewalls through failure testing.